Universiteit Leiden
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Relaxed SEM-based out-of-sample predictions

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Explanation and prediction

$\boldsymbol{x_{i,1}}$	$\boldsymbol{x_{i,2}}$	$\boldsymbol{x_{i,3}}$	$\boldsymbol{y_{i,1}}$	$\boldsymbol{y_{i,2}}$
$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	$y_{1,1}$	$y_{1,2}$
$x_{2,1}$	$x_{2,2}$	$x_{2,3}$	$y_{2,1}$	$y_{2,2}$
$x_{3,1}$	$x_{3,2}$	$x_{3,3}$	$y_{3,1}$	$y_{3,2}$
$x_{4,1}$	$x_{4,2}$	$x_{4,3}$	$y_{4,1}$	$y_{4,2}$

Explanation

$$\mathbf{Y} = f(\mathbf{X}) + \mathbf{E}$$

Explanation and prediction

$\boldsymbol{x}_{i,1}$	$\boldsymbol{x}_{i,2}$	$\boldsymbol{x}_{i,3}$	$\boldsymbol{y}_{i,1}$	$\boldsymbol{y}_{i,2}$
$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	$y_{1,1}$	$y_{1,2}$
$x_{2,1}$	$x_{2,2}$	$x_{2,3}$	$y_{2,1}$	$y_{2,2}$
$x_{3,1}$	$x_{3,2}$	$x_{3,3}$	$y_{3,1}$	$y_{3,2}$
$x_{4,1}$	$x_{4,2}$	$x_{4,3}$	$y_{4,1}$	$y_{4,2}$
$x_{5,1}$	$x_{5,2}$	$x_{5,3}$	?	?
$x_{6,1}$	$x_{6,2}$	$x_{6,3}$	?	?

$\mathbf{X}_0$ (rows 5–6, columns x) $\quad \hat{\mathbf{Y}}$ (rows 5–6, columns y)

Explanation

$$\mathbf{Y} = f(\mathbf{X}) + \mathbf{E}$$

Prediction

$$\hat{\mathbf{Y}} = \hat{f}(\mathbf{X}_0)$$

Explanation and prediction

$\boldsymbol{x_{i,1}}$	$\boldsymbol{x_{i,2}}$	$\boldsymbol{x_{i,3}}$	$\boldsymbol{y_{i,1}}$	$\boldsymbol{y_{i,2}}$
$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	$y_{1,1}$	$y_{1,2}$
$x_{2,1}$	$x_{2,2}$	$x_{2,3}$	$y_{2,1}$	$y_{2,2}$
$x_{3,1}$	$x_{3,2}$	$x_{3,3}$	$y_{3,1}$	$y_{3,2}$
$x_{4,1}$	$x_{4,2}$	$x_{4,3}$	$y_{4,1}$	$y_{4,2}$
$x_{5,1}$	$x_{5,2}$	$x_{5,3}$	?	?
$x_{6,1}$	$x_{6,2}$	$x_{6,3}$	?	?

$\mathbf{X}_0$

$\hat{\mathbf{Y}}$

OLS regression

$$\mathbf{Y} = \mathbf{1}_N \bar{\mathbf{y}}^\top + (\mathbf{X} - \mathbf{1}_N \bar{\mathbf{x}}^\top)\mathbf{S}_{XX}^{-1}\mathbf{S}_{XY} + \mathbf{E}$$

OLS prediction

$$\hat{\mathbf{Y}} = \mathbf{1}_N \bar{\mathbf{y}}^\top + (\mathbf{X}_0 - \mathbf{1}_N \bar{\mathbf{x}}^\top)\mathbf{S}_{XX}^{-1}\mathbf{S}_{XY}$$

Explanation and prediction

$x_{i,1}$	$x_{i,2}$	$x_{i,3}$	$y_{i,1}$	$y_{i,2}$
$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	$y_{1,1}$	$y_{1,2}$
$x_{2,1}$	$x_{2,2}$	$x_{2,3}$	$y_{2,1}$	$y_{2,2}$
$x_{3,1}$	$x_{3,2}$	$x_{3,3}$	$y_{3,1}$	$y_{3,2}$
$x_{4,1}$	$x_{4,2}$	$x_{4,3}$	$y_{4,1}$	$y_{4,2}$
$x_{5,1}$	$x_{5,2}$	$x_{5,3}$	?	?
$x_{6,1}$	$x_{6,2}$	$x_{6,3}$	?	?

$\mathbf{X}_0$ $\hat{\mathbf{Y}}$

OLS prediction

$$\hat{\mathbf{Y}} = \mathbf{1}_N \bar{\mathbf{y}}^\top + (\mathbf{X}_0 - \mathbf{1}_N \bar{\mathbf{x}}^\top)\mathbf{S}_{XX}^{-1}\mathbf{S}_{XY}$$

Limitations of regression

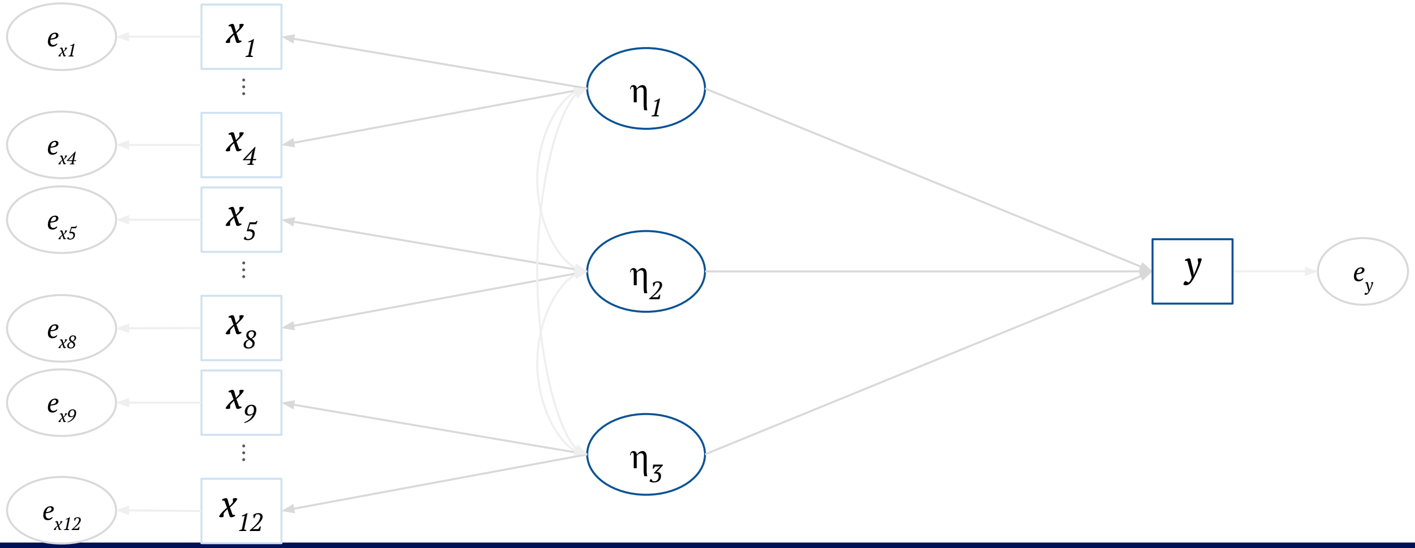
- Measurement models of complex constructs cannot be easily incorporated — **such constructs are common in psychological research**
- Unrestricted estimation of all effects — **inflexible to incorporate complex theories**

Structural equation models (SEMs)

Combination of **measurement and structural (regression) models**

Flexible model building framework

Restricted model-implied covariance matrix estimation based on prior theory or model modifications



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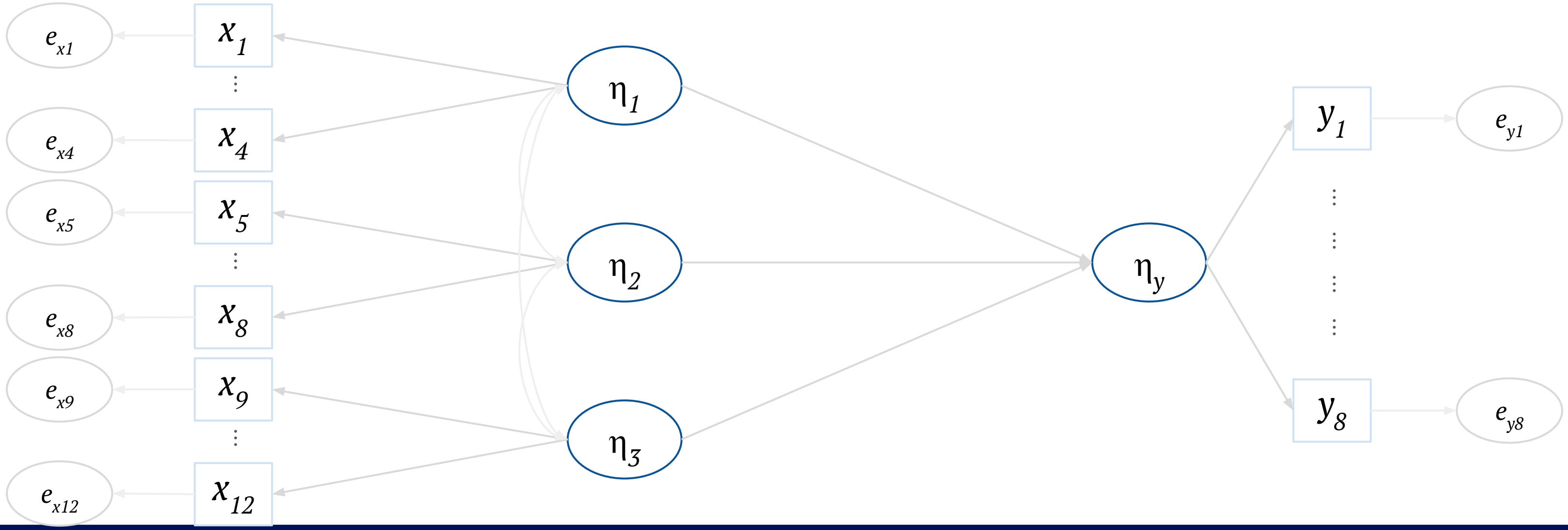
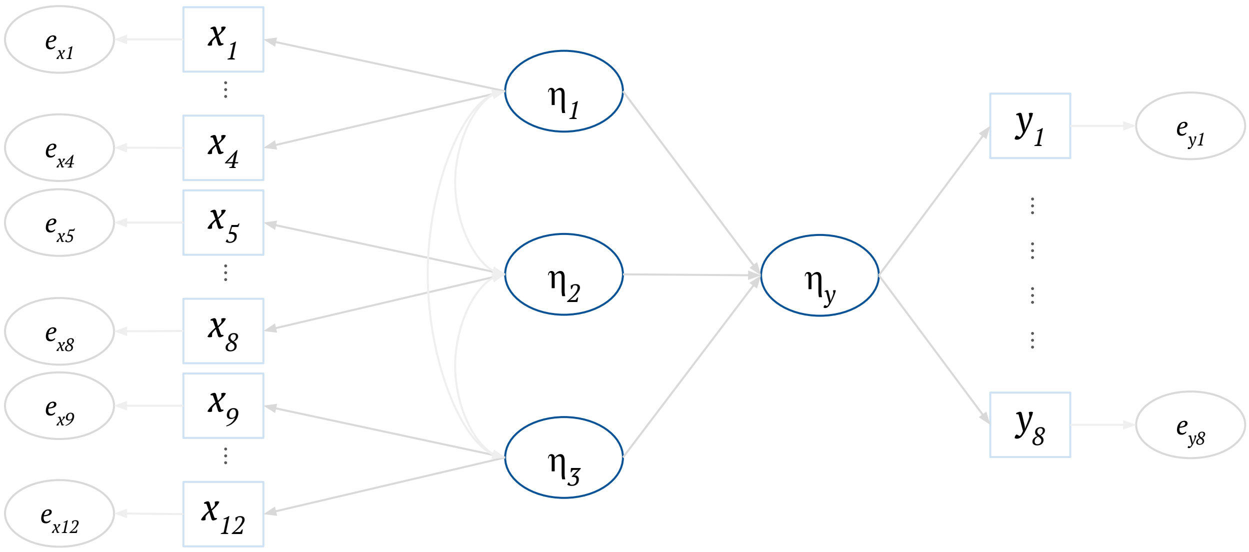


Estimating SEMs

$\boldsymbol{x_{i,1}}$	…	$\boldsymbol{x_{i,12}}$	$\boldsymbol{y_{i,1}}$	…	$\boldsymbol{y_{i,8}}$
$x_{1,1}$	…	$x_{1,12}$	$y_{1,1}$	…	$y_{1,8}$
$x_{2,1}$	…	$x_{2,12}$	$y_{2,1}$	…	$y_{2,8}$
$x_{3,1}$	…	$x_{3,12}$	$y_{3,1}$	…	$y_{3,8}$
$x_{4,1}$	…	$x_{4,12}$	$y_{4,1}$	…	$y_{4,8}$



$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix}$$

Model-implied mean vector

$$\hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \hat{\boldsymbol{\Sigma}}_{XY} \\ \hat{\boldsymbol{\Sigma}}_{YX} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix}$$

Model-implied covariance matrix

SEM-based prediction

	$x_{i,1}$	…	$x_{i,12}$	$y_{i,1}$	…	$y_{i,8}$	
	$x_{1,1}$	…	$x_{1,12}$	$y_{1,1}$	…	$y_{1,8}$	
	$x_{2,1}$	…	$x_{2,12}$	$y_{2,1}$	…	$y_{2,8}$	
	$x_{3,1}$	…	$x_{3,12}$	$y_{3,1}$	…	$y_{3,8}$	
	$x_{4,1}$	…	$x_{4,12}$	$y_{4,1}$	…	$y_{4,8}$	
	$x_{5,1}$	…	$x_{5,12}$	?	…	?	
$\mathbf{X}_0$	$x_{6,1}$	…	$x_{6,12}$	?	…	?	$\hat{\mathbf{Y}}$
	$x_{7,1}$	…	$x_{7,12}$	?	…	?	

SEM-based prediction (De Rooij et al., 2022)

$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top)\hat{\boldsymbol{\Sigma}}_{XX}^{-1}\hat{\boldsymbol{\Sigma}}_{XY}$$

- Outperformed regression-based prediction
- Robust against non-normality
- Sensitive to model misspecification (addition of direct effects between *x* and *y* in the data-generating model)

De Rooij, M., Karch, J. D., Fokkema, M., Bakk, Z., Pratiwi, B. C., & Kelderman, H. (2022). SEM-based out-of-sample predictions. *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(1), 132–148. https://doi.org/10.1080/10705511.2022.2061494

SEM-based prediction

$x_{i,1}$	…	$x_{i,12}$	$y_{i,1}$	…	$y_{i,8}$
$x_{1,1}$	…	$x_{1,12}$	$y_{1,1}$	…	$y_{1,8}$
$x_{2,1}$	…	$x_{2,12}$	$y_{2,1}$	…	$y_{2,8}$
$x_{3,1}$	…	$x_{3,12}$	$y_{3,1}$	…	$y_{3,8}$
$x_{4,1}$	…	$x_{4,12}$	$y_{4,1}$	…	$y_{4,8}$
$x_{5,1}$	…	$x_{5,12}$	?	…	?
$x_{6,1}$	…	$x_{6,12}$	?	…	?
$x_{7,1}$	…	$x_{7,12}$	?	…	?

$\mathbf{X}_0$ $\hat{\mathbf{Y}}$

SEM-based prediction (De Rooij et al., 2022)

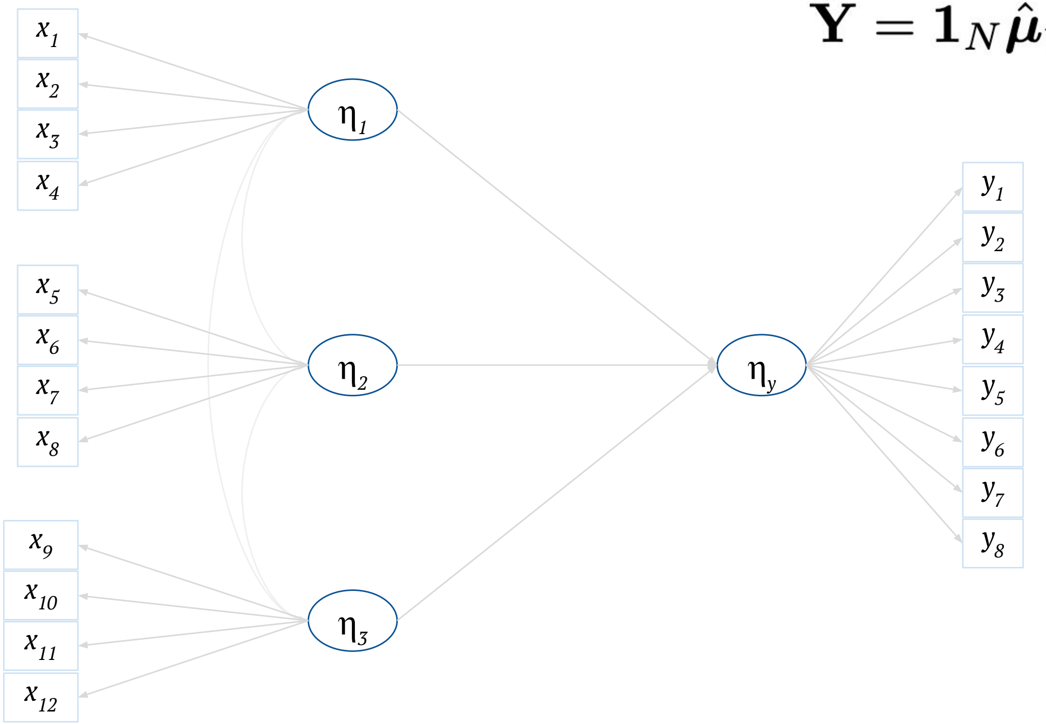
$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top)\hat{\boldsymbol{\Sigma}}_{XX}^{-1}\hat{\boldsymbol{\Sigma}}_{XY}$$

- Sensitive to model misspecification (addition of direct effects between *x* and *y* in the data-generating model)

Biased $\hat{\Sigma}$, and thus, inaccurate $\hat{\text{Y}}$ values

De Rooij, M., Karch, J. D., Fokkema, M., Bakk, Z., Pratiwi, B. C., & Kelderman, H. (2022). SEM-based out-of-sample predictions. *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(1), 132–148. https://doi.org/10.1080/10705511.2022.2061494

A (mis)specified SEM



$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top)\hat{\boldsymbol{\Sigma}}_{XX}^{-1}\hat{\boldsymbol{\Sigma}}_{XY}$$

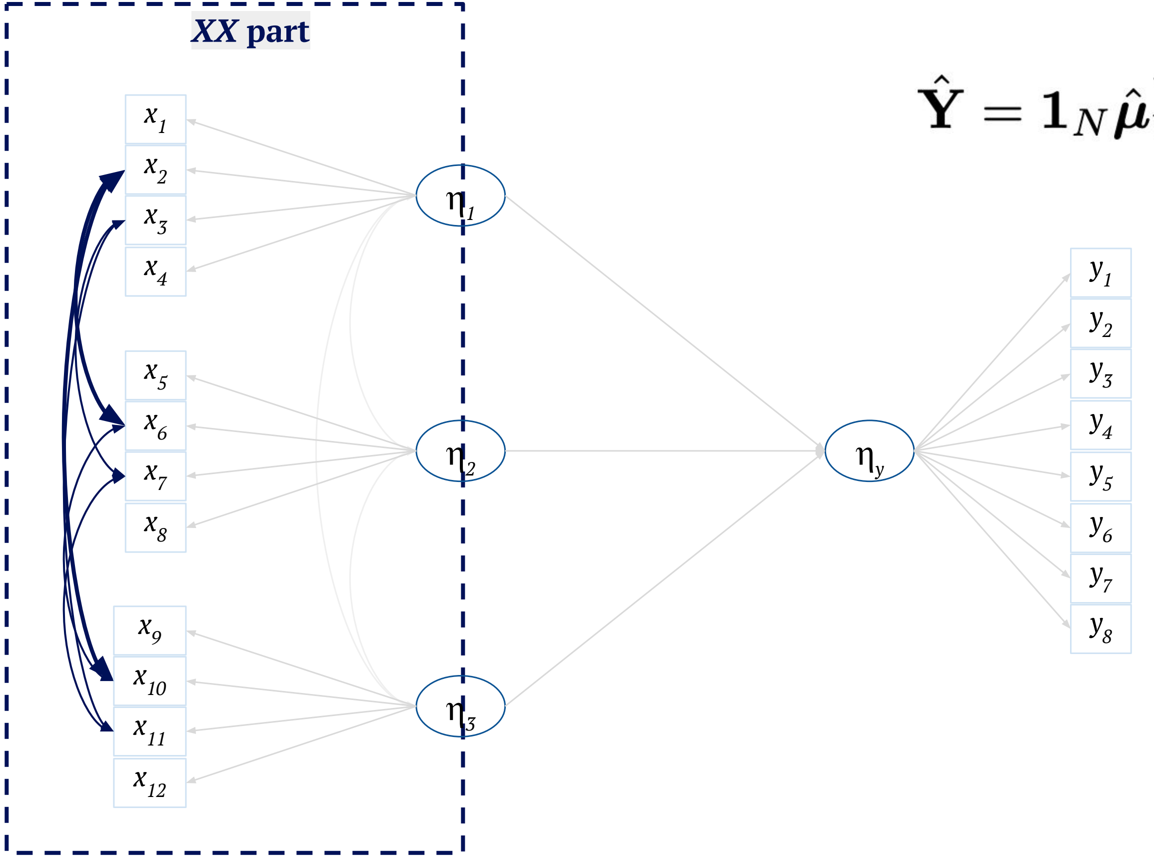
$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix}$$

Model-implied mean vector

$$\hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \hat{\boldsymbol{\Sigma}}_{XY} \\ \hat{\boldsymbol{\Sigma}}_{YX} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix}$$

Model-implied covariance matrix

A (mis)specified SEM



$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top)\underline{\hat{\boldsymbol{\Sigma}}_{XX}^{-1}\hat{\boldsymbol{\Sigma}}_{XY}}$$

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix}$$

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Model-implied covariance matrix

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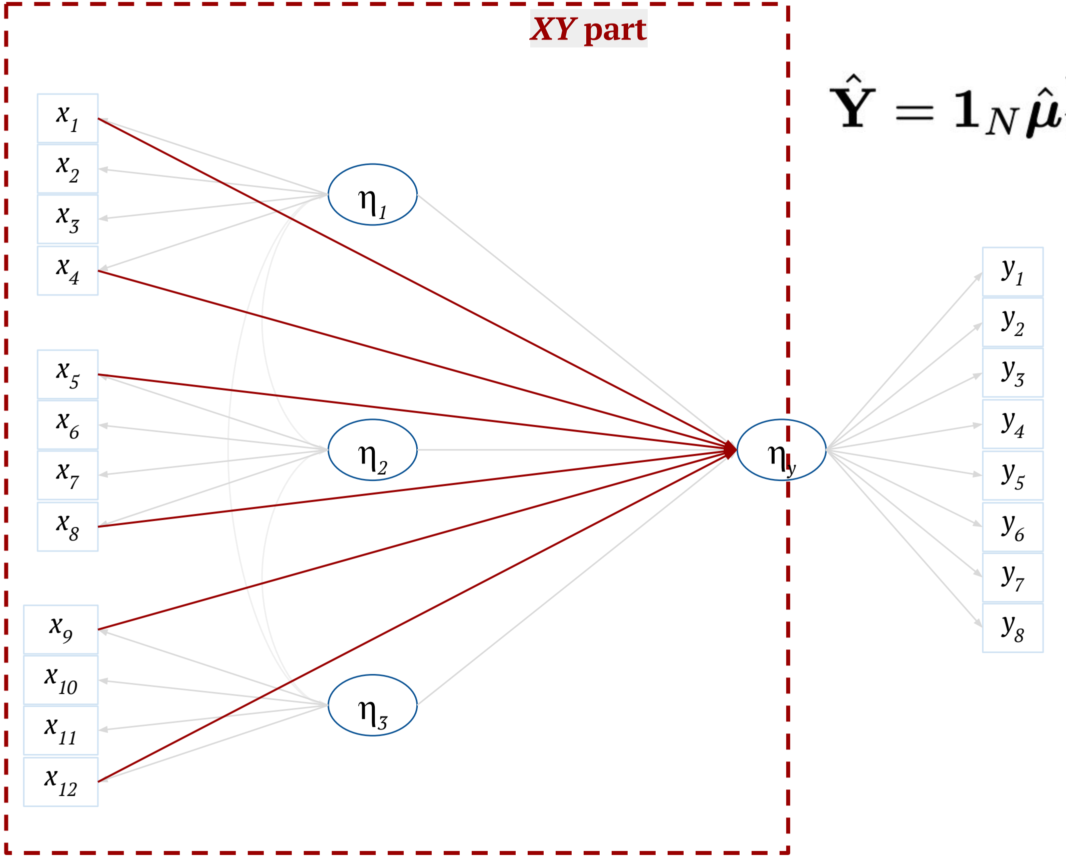


$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top)\hat{\boldsymbol{\Sigma}}_{XX}^{-1} \underline{\hat{\boldsymbol{\Sigma}}_{XY}}$$

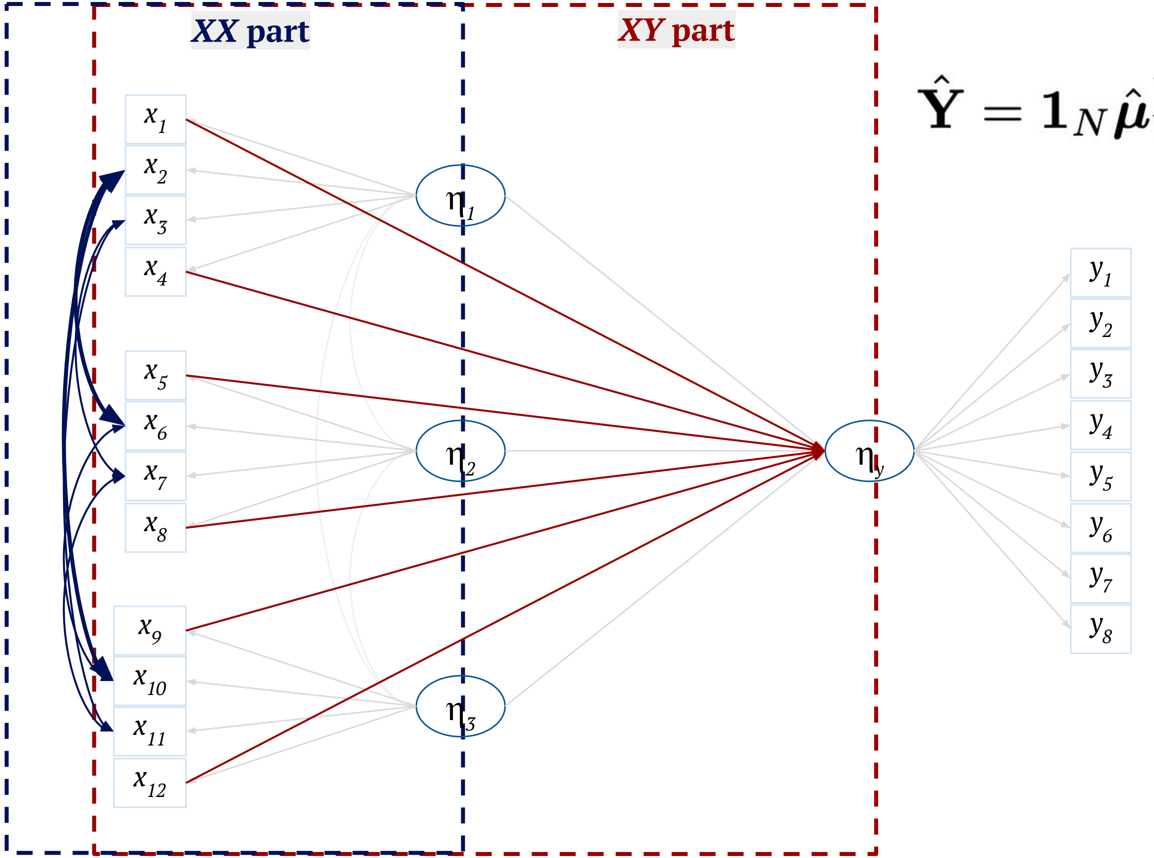
$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix}$$

Model-implied mean vector

$$\hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \boxed{\hat{\boldsymbol{\Sigma}}_{XY}} \\ \boxed{\hat{\boldsymbol{\Sigma}}_{YX}} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix}$$

Model-implied covariance matrix

A (mis)specified SEM



$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top)\underline{\hat{\boldsymbol{\Sigma}}_{XX}^{-1}\underline{\hat{\boldsymbol{\Sigma}}_{XY}}}$$

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix}$$

Model-implied mean vector

$$\hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \hat{\boldsymbol{\Sigma}}_{XY} \\ \hat{\boldsymbol{\Sigma}}_{YX} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix}$$

Model-implied covariance matrix

A solution: Relaxed SEM-based prediction

$\boldsymbol{x_{i,1}}$	…	$\boldsymbol{x_{i,12}}$	$\boldsymbol{y_{i,1}}$	…	$\boldsymbol{y_{i,8}}$
$x_{1,1}$	…	$x_{1,12}$	$y_{1,1}$	…	$y_{1,8}$
$x_{2,1}$	…	$x_{2,12}$	$y_{2,1}$	…	$y_{2,8}$
$x_{3,1}$	…	$x_{3,12}$	$y_{3,1}$	…	$y_{3,8}$
$x_{4,1}$	…	$x_{4,12}$	$y_{4,1}$	…	$y_{4,8}$
$x_{5,1}$	…	$x_{5,12}$	?	…	?
$x_{6,1}$	…	$x_{6,12}$	?	…	?
$x_{7,1}$	…	$x_{7,12}$	?	…	?

$\mathbf{X}_0$ $\hat{\mathbf{Y}}$

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix}$$

Model-implied mean vector

$$\hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \hat{\boldsymbol{\Sigma}}_{XY} \\ \hat{\boldsymbol{\Sigma}}_{YX} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix}$$

Model-implied covariance matrix

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & \mathbf{S}_{YY} \end{bmatrix}$$

Sample covariance matrix

A solution: Relaxed SEM-based prediction

$x_{i,1}$	…	$x_{i,12}$	$y_{i,1}$	…	$y_{i,8}$
$x_{1,1}$	…	$x_{1,12}$	$y_{1,1}$	…	$y_{1,8}$
$x_{2,1}$	…	$x_{2,12}$	$y_{2,1}$	…	$y_{2,8}$
$x_{3,1}$	…	$x_{3,12}$	$y_{3,1}$	…	$y_{3,8}$
$x_{4,1}$	…	$x_{4,12}$	$y_{4,1}$	…	$y_{4,8}$
$x_{5,1}$	…	$x_{5,12}$	?	…	?
$x_{6,1}$	…	$x_{6,12}$	?	…	?
$x_{7,1}$	…	$x_{7,12}$	?	…	?

$\mathbf{X}_0$ $\hat{\mathbf{Y}}$

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix} \quad \hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \hat{\boldsymbol{\Sigma}}_{XY} \\ \hat{\boldsymbol{\Sigma}}_{YX} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix} \quad \mathbf{S} = \begin{bmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & \mathbf{S}_{YY} \end{bmatrix}$$

$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top) \Big(\underbrace{(1-\gamma_1)\hat{\boldsymbol{\Sigma}}_{XX} + \gamma_1 \mathbf{S}_{XX}}_{\textit{XX}\text{ part}} \Big)^{-1} \Big(\underbrace{(1-\gamma_2)\hat{\boldsymbol{\Sigma}}_{XY} + \gamma_2 \mathbf{S}_{XY}}_{\textit{XY}\text{ part}} \Big)$$

A solution: Relaxed SEM-based prediction

$\boldsymbol{x_{i,1}}$	...	$\boldsymbol{x_{i,12}}$	$\boldsymbol{y_{i,1}}$	...	$\boldsymbol{y_{i,8}}$
$x_{1,1}$	...	$x_{1,12}$	$y_{1,1}$	...	$y_{1,8}$
$x_{2,1}$	...	$x_{2,12}$	$y_{2,1}$	...	$y_{2,8}$
$x_{3,1}$	...	$x_{3,12}$	$y_{3,1}$	...	$y_{3,8}$
$x_{4,1}$	...	$x_{4,12}$	$y_{4,1}$	...	$y_{4,8}$
$x_{5,1}$	...	$x_{5,12}$	?	...	?
$x_{6,1}$	...	$x_{6,12}$	?	...	?
$x_{7,1}$	...	$x_{7,12}$	?	...	?

$\mathbf{X}_0$ $\hat{\mathbf{Y}}$

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \hat{\boldsymbol{\mu}}_X \\ \hat{\boldsymbol{\mu}}_Y \end{bmatrix} \quad \hat{\boldsymbol{\Sigma}} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{XX} & \hat{\boldsymbol{\Sigma}}_{XY} \\ \hat{\boldsymbol{\Sigma}}_{YX} & \hat{\boldsymbol{\Sigma}}_{YY} \end{bmatrix} \quad \mathbf{S} = \begin{bmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & \mathbf{S}_{YY} \end{bmatrix}$$

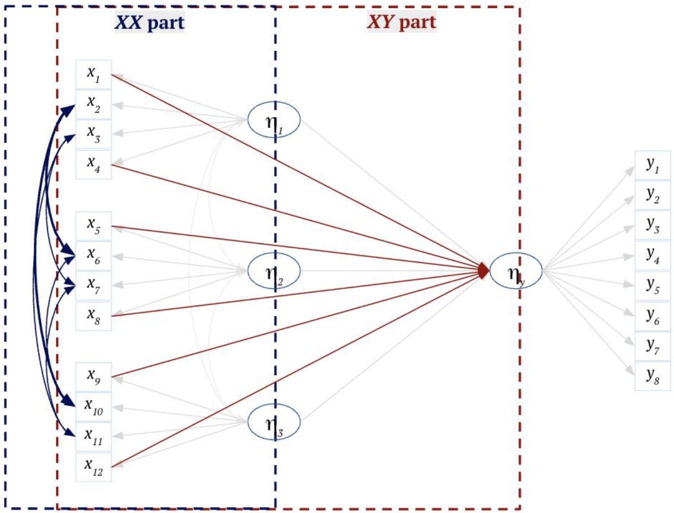
$$\hat{\mathbf{Y}} = \mathbf{1}_N \hat{\boldsymbol{\mu}}_Y^\top + (\mathbf{X}_0 - \mathbf{1}_N \hat{\boldsymbol{\mu}}_X^\top) \Big(\underbrace{(1-\gamma_1)\hat{\boldsymbol{\Sigma}}_{XX} + \gamma_1 \mathbf{S}_{XX}}_{\textit{XX}\text{ part}} \Big)^{-1} \Big(\underbrace{(1-\gamma_2)\hat{\boldsymbol{\Sigma}}_{XY} + \gamma_2 \mathbf{S}_{XY}}_{\textit{XY}\text{ part}} \Big)$$

$0 \leqslant \{\gamma_1, \gamma_2\} \leqslant 1$ chosen by k-fold cross-validation

$\{\gamma_1, \gamma_2\}$ **< 0.5** Favour $\hat{\Sigma}$ (mis)specified SEM

$\{\gamma_1, \gamma_2\}$ **> 0.5** Favour $\boldsymbol{S}$ (saturated model/OLS)

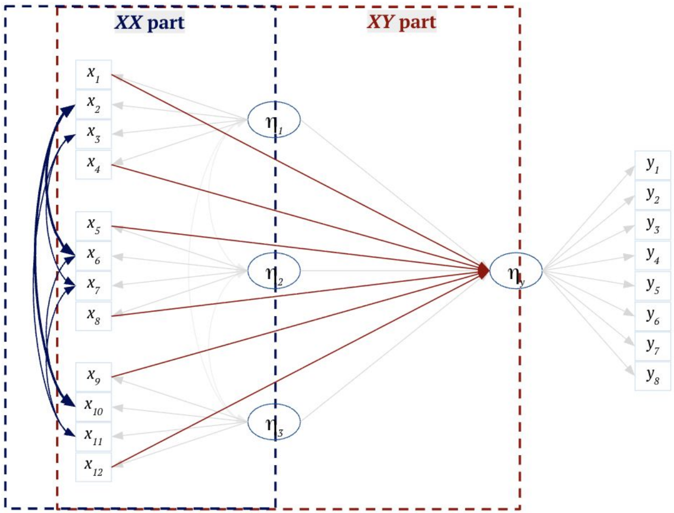
Simulation



Misspecified both the *XX* and *XY* parts of the model

- 500 replications
- Selecting γ_1 and γ_2
  - Calibration set of N_{cal} = 100, 250, 500, or 1000
  - *10*-fold cross-validation
- Prediction
  - Prediction set of N_{pred} = 10,000

Simulation



Misspecified both the *XX* and *XY* parts of the model

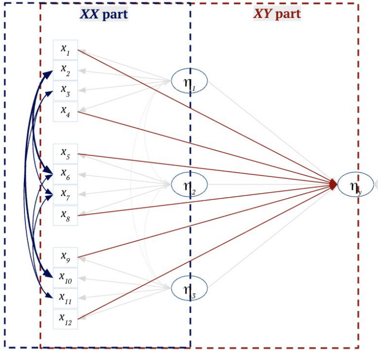
- Compare four methods:
  - SEM prediction (De Rooij et al., 2022)
  - **Relaxed SEM prediction**
  - OLS prediction
  - Elastic net prediction

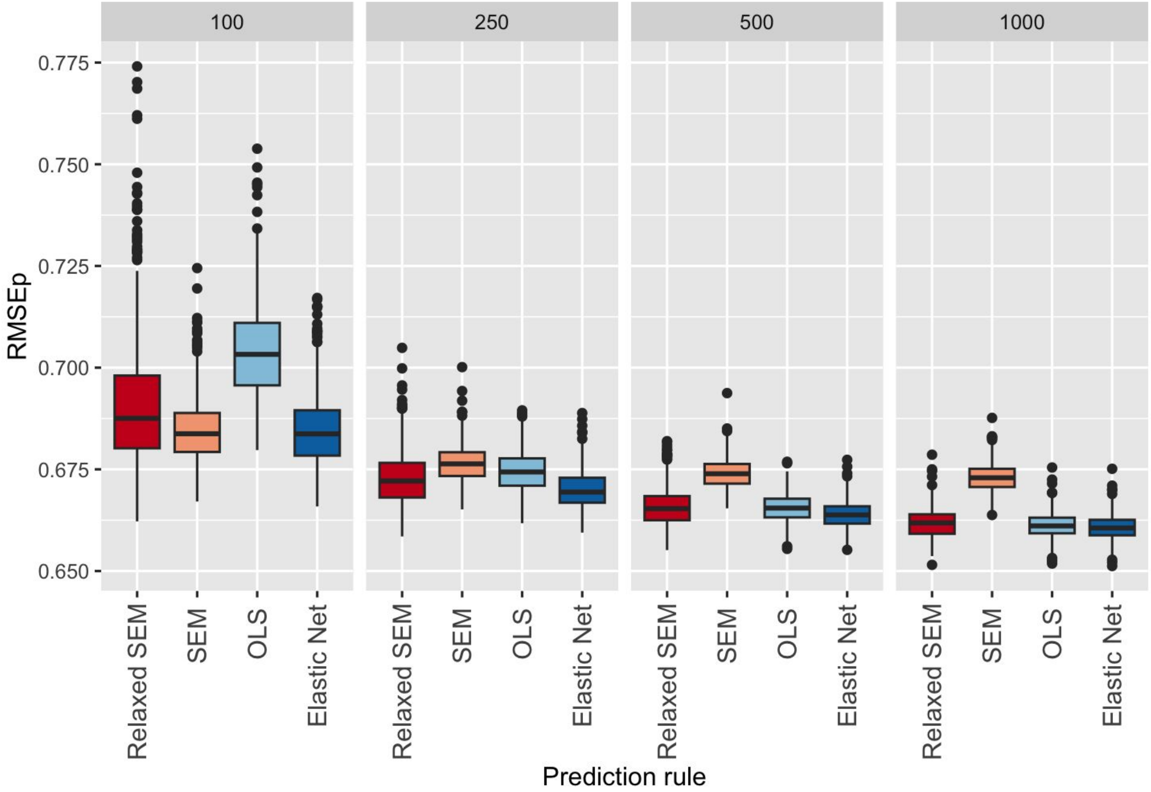
- **RMSEp** as a measure of prediction error

$$\mathrm{RMSEp} = \sqrt{\frac{1}{n \times R}\sum_{i=1}^{n}\sum_{r=1}^{R}(y_{ir} - \hat{y}_{ir})^2}$$

where $n = N_{pred}$ and R = number of y variables (R = 8)

Results: A weakly misspecified model

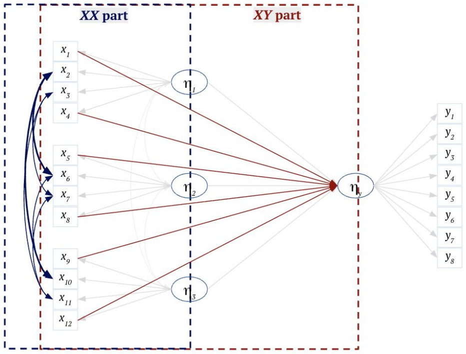


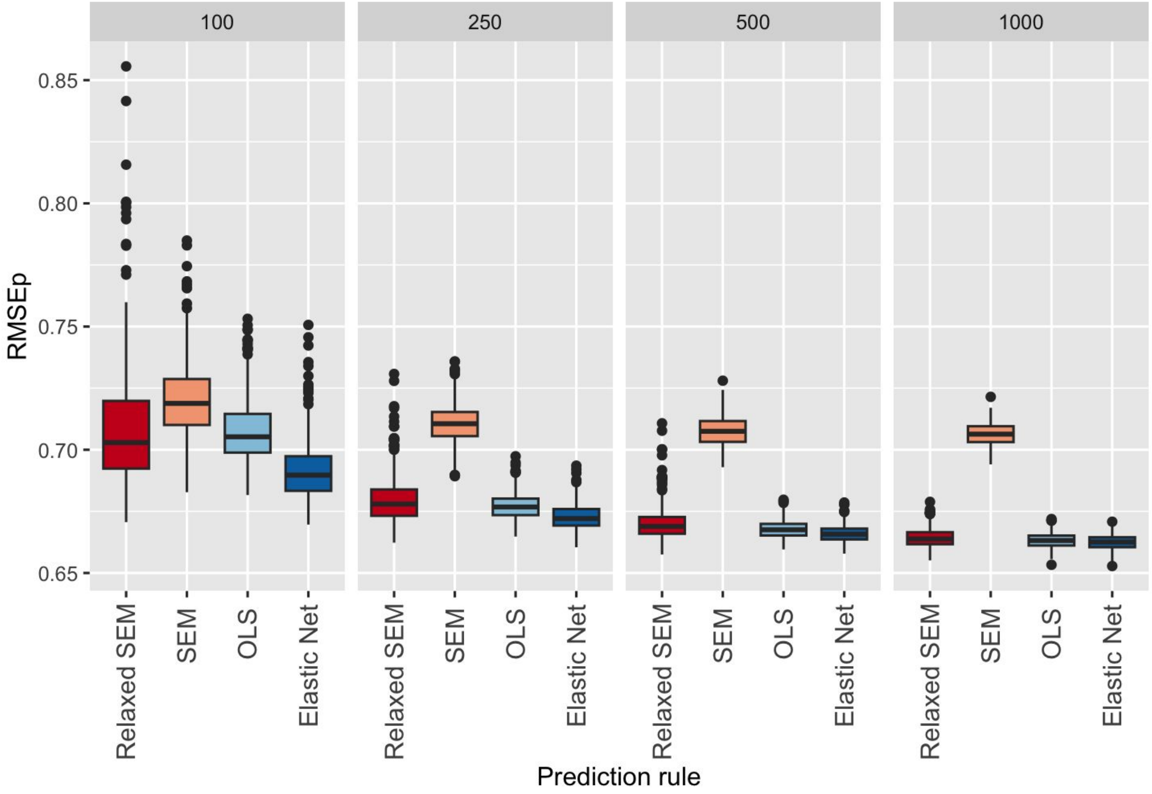


Some observations

- In small samples
  - SEM outperforms relaxed SEM on account of the bias-variance tradeoff
  - Outliers— due to what is called the $N \times (P + R)$ ratio (models will generalise poorly in small samples as the calibration data does not provide sufficient information to accurately estimate the model).
- Relaxed SEM is asymptotically comparable to elastic net

Results: A strongly misspecified model





Some observations

- In small samples
  - Relaxed SEM outperforms SEM as the misspecification is much stronger and more complex models are preferred for prediction despite the small calibration size.
  - Outliers— due to what is called the $N \times (P + R)$ ratio; instability of hyperparameter optimisation
- Relaxed SEM is asymptotically comparable to elastic net

Conclusions and next steps

- Relaxed SEM-based prediction shows promise in addressing model misspecification in predictive contexts
- But this is only a first implementation!
  - Possible to compute hyperparameters analytically (Frobenius norm discrepancies)
  - Use different shrinkage targets based on the type of misspecification
- For SEM-based prediction in general,
  - Develop rules for ordinal, mixed, and clustered data
  - Integrate explanation and prediction to build more generalisable models

Questions?

Or, feel free to reach out: **a.m.bhangale@fsw.leidenuniv.nl**